



CODE SPORT

In this month's column, we discuss natural language processing and its role in information retrieval.

In last month's column, we had discussed various compression schemes for maintaining the data structures associated with information retrieval systems. You will recall that the major tasks of the information retrieval (IR) systems are to process the user query, find documents that are potentially relevant to the user query and return a ranked result containing the retrieved documents. In other words, IR can be thought of as three major processes—query processing, matching (the query to the relevant documents in the collection) and ranking the results using certain heuristics so that the documents most relevant to the user are listed earlier in the results.

Earlier, we had discussed how user queries can be processed using tokenisation, stemming and lemmatisation techniques. For matching, we had discussed techniques such as the Boolean model of information retrieval, which performs the binary classification of documents as either relevant or irrelevant to the query; and the vector space model which assigns weightage with respect to the relevance of the document to the query under consideration. Ranking of retrieved results can be based on the weights associated with relevance, or through external techniques like page ranking, which ranks the results depending on how many authoritative page links there are to a particular page, etc.

While we described the complete cycle of information retrieval in terms of query processing, matching and ranking, we had not really used any semantic information contained either in the query or in the document itself to help the information retrieval process. Statistical properties such as Term Frequency-

Inverse Document Frequency (TF-IDF) associated with terms in the document have been successfully used to build functional IR systems. However, such statistical techniques have their own drawbacks. For instance, if we compare how human beings would perform information retrieval techniques with automatic machine driven information retrieval, it is clear that human beings are more comfortable analysing the syntactic and semantic contents of the documents/queries to aid information retrieval. As an example, consider a simple query, "What are the foods that do not contain magnesium?" As human beings, we all know that we are looking for foods that do not contain magnesium. But try typing this query in Google, or any other search engine. You will find that most of them ignore the negation term and return you results featuring 'foods that contain magnesium,' 'whether eating magnesium is healthy,' 'how you can get enough magnesium,' etc?" The results returned by the search engine are absolutely useless to users because they are the very opposite of their information needs. If the same query is asked to a human being, such a mistake will never happen because the semantic meaning of the query is well understood by humans and, hence, they will only give details on which foods do not contain magnesium. Just doing a word matching or shallow parsing will not suffice in this case, since the word 'NOT' being a common word, it is typically dropped during the query processing itself. On the other hand, if natural language processing techniques are applied to query processing, such a mistake will